ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

ELECTRONICS AND DIGITAL CONCEPTS FOR ROBOTICS

PAPER 1 · PRACTICE C · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 3 Electronics and Digital Concepts for Robotics paper 1 · practice c.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 3. It is not a Department of Higher Education and Training marking guideline.

How to mark: each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Use consistent accuracy. Accept other correct answers and neat, recognisable diagrams.

SECTION A

QUESTION 1

1.1 B ✓
1.2 A ✓
1.3 C ✓
1.4 D ✓
1.5 A ✓
1.6 B ✓
1.7 C ✓
1.8 B ✓
1.9 A ✓
1.10 D ✓

(10)

[10]

QUESTION 2

2.1 True ✓
2.2 False ✓
2.3 True ✓
2.4 False ✓
2.5 True ✓
2.6 False ✓
2.7 True ✓
2.8 True ✓
2.9 False ✓
2.10 True ✓

(10)

[10]

QUESTION 3

3.1 C ✓
3.2 F ✓
3.3 I ✓
3.4 H ✓
3.5 K ✓

3.6 B ✓

3.7 D ✓

3.8 A ✓

3.9 G ✓

3.10 J ✓

(10)

[10]

TOTAL SECTION A: [30]

SECTION B

QUESTION 4

4.1.1 The total number of magnetic lines of force in a magnetic field (unit: weber) ✓ (1)

4.1.2 The amount of magnetic flux passing through one square metre at right angles to the flux (unit: tesla) ✓ (1)

4.1.3 How easily a material allows magnetic flux to pass through it ✓ (1)

4.1.4 A coil wound on a soft-iron core that becomes a magnet only while current flows through it ✓ (1)

4.2.1 $\text{mmf} = \text{N} \times \text{I} = 400 \times 2{,}5 \checkmark = 1\,000\ \text{At} \checkmark$ (2)

4.2.2 $\text{H} = \text{mmf} \div \text{l} = 1\,000 \div 0{,}2 \checkmark = 5\,000\ \text{At/m} \checkmark$ (2)

4.3 First law: an emf is induced in a conductor whenever it cuts magnetic lines of force, or the flux linking a coil changes ✓✓. Second law: the size of the induced emf is proportional to the rate at which the flux is cut or changes ✓✓. (4)

4.4 $\text{e} = \text{B} \times \text{l} \times \text{v} \checkmark = 0{,}8 \times 0{,}3 \times 5 \checkmark = 1{,}2\ \text{V} \checkmark$ (3)

[15]

QUESTION 5

5.1.1 $1 \div \text{R}_\text{P} = 1 \div 10 + 1 \div 15 + 1 \div 30 \checkmark = 6 \div 30 \checkmark; \text{R}_\text{P} = 5\ \Omega \checkmark$ (3)

5.1.2 $\text{R}_\text{T} = 5 + 5 = 10\ \Omega \checkmark$ (1)

5.1.3 $\text{I} = \text{V} \div \text{R} = 24 \div 10 \checkmark = 2{,}4\ \text{A} \checkmark$ (2)

5.1.4 $\text{V} = \text{I} \times \text{R} = 2{,}4 \times 5 \checkmark = 12\ \text{V} \checkmark$ (2)

5.2.1 $\text{N}_\text{s} = \text{N}_\text{p} \times \text{V}_\text{s} \div \text{V}_\text{p} \checkmark = 1\,150 \times 12 \div 230 \checkmark = 60\ \text{turns} \checkmark$ (3)

5.2.2 $\text{I}_\text{p} = \text{I}_\text{s} \times \text{V}_\text{s} \div \text{V}_\text{p} = 2 \times 12 \div 230 \checkmark = 0{,}10\ \text{A} \checkmark$ (2)

5.3 Any FOUR ✓✓✓✓: yoke (frame); field poles and field windings; armature (core and windings); commutator; brushes; shaft; bearings; end shields. (4)

5.4 In a series motor the field winding is connected in series with the armature ✓; in a shunt motor the field winding is connected in parallel (across) the armature ✓. (2)

5.5 Any ONE ✓: electric trains / trams; cranes and hoists; starter motors in vehicles; electric drills – where a high starting torque is needed. (1)

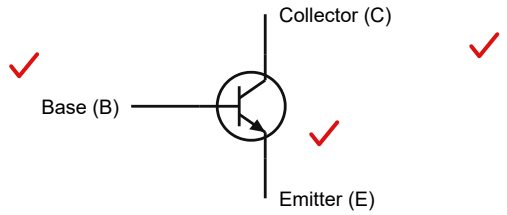
[20]

QUESTION 6

6.1 $R = \rho L \div A \checkmark = 0{,}028 \times 10^{-6} \times 500 \div (2{,}5 \times 10^{-6}) \checkmark\checkmark = 5{,}6\ \Omega \checkmark$ (4)

6.2 $R_t = R_0(1 + \alpha_0 t) \checkmark = 20(1 + 0{,}00427 \times 60) \checkmark = 25{,}12\ \Omega \checkmark$ (3)

6.3



(Correct symbol with the emitter arrow pointing outwards 1, base labelled 1, collector and emitter labelled 1)

(3)

6.4 The Arduino pin is connected through a base resistor to the base of the transistor ✓; when the pin goes HIGH, base current flows and the transistor saturates (switches fully on), so current flows from the 12 V supply through the relay coil to ground ✓; when the pin goes LOW, the transistor cuts off and the relay switches off (a diode across the coil protects the transistor) ✓. (3)

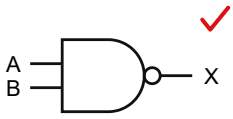
6.5 $I_E = I_C + I_B$ ✓✓ (2)

[15]

QUESTION 7

7.1 Any SIX in a logical order ✓✓✓✓✓✓: switch off the circuit; set the meter to the correct current range (start on the highest range / A or mA); put the red lead in the correct current socket and the black lead in COM; break the circuit where the current is to be measured; connect the meter in series so all the current flows through it; switch on and read the value; switch off, remove the meter and reconnect the circuit; return the leads and selector to the voltage position. (6)

7.2



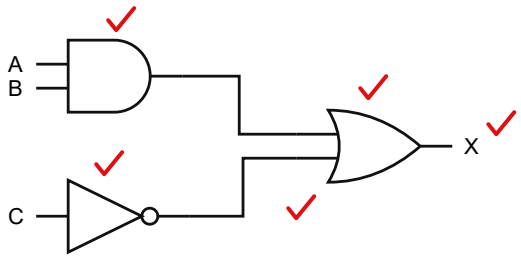
(Correct NAND symbol: AND shape with a small circle at the output 1; truth table below 4)

(5)

A	B	X
0	0	1

0	1	1
1	0	1
1	1	0

7.3



(Two-input AND gate with inputs A and B 1, NOT gate on input C 1, two-input OR gate 1 with the AND output and the NOT output as its inputs 1, output labelled X 1)

(5)

7.4 Any FOUR ✓✓✓✓: keep it clean and dry (wipe with a dry or slightly damp cloth, not a wet one); store it in its case; remove the batteries when it will not be used for a long time; switch it off after use; check the leads for damage before use; do not drop it or expose it to heat; have it calibrated regularly; use the correct range. (4)

[20]

TOTAL SECTION B: [70]

TOTAL: 100

END OF MARKING GUIDELINE